

National College of Computer Studies
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Bsc. CSIT

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Documentation Report on: Nutrition Tracker System

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Acknowledgement

I want to use this chance to express my thanks to my instructor for providing me with useful guidance during the designing and development of this project. It is due to the constructive criticism from my instructor that this report has come into existence.

I want to thank all my fellow students who assisted me in testing the Nutrition Tracker software application. This project has been based on understanding the basics of Java Desktop Application, MySQL Connectivity, DAO architecture, and Swing UI design.

Abstract

The Nutrition Tracker is a Java desktop application that helps people keep track of their daily food intake and nutritional goals. The system has two roles: normal user and administrator. Users can save a personal food database, track meals eaten each day, view their daily nutrition summary versus personalized goals, and search their entire food history. Administrators can view users' food logs, send notes/dietary guidance directly to registered users and manage user accounts.

This documentation describes the software analysis and design of the Nutrition Tracker system. It describes the project objectives and scope, the background and existing systems, the overall description of the system including product features and user classes, the external interface requirements, database schema, system architecture and class structure and non-functional requirements.

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1. Introduction

Nutrition Tracker is a desktop application, used by a user to track the consumed foods throughout the day and record progress made towards their specific nutritional targets. The system has a structured view that allows the user to create their own food entries including their nutritional values, log meals specifying exact gram values, and viewing nutrition summary for any particular date. The system enables administrators to maintain users and send dietary notes.

1.1. Purpose

This project aims to be an approachable, usable, desktop application tool for people to keep track of their eating. This alleviates dependence on outside spreadsheet documents or manually kept records, as a single database-driven application, with role-based authentication, will be the foundation. The scope ranges from personal health tracking to the ability for a trainer/nutritionist to take a very light look at administrative oversight.

1.2. Project Scope

The scope of the Nutrition Tracker includes:

- User registration and role-based login (user and admin).
- Personal food database management (add, edit, delete food entries).
- Daily food logging with gram-based consumption tracking.
- Nutrition summary calculation and goal comparison.
- Food history browsing by date.
- Admin: user account management and notes dispatch.

1.3. Document Convention

This document follows a Software Analysis and Design report format. The main text uses Times New Roman, 12 pt font, with clear numbered headings and aligned page numbers in the footer. Diagrams are captioned and placed in the Other Requirements section.

1.4. Additional Information

The system is implemented in Java with Swing GUI toolkit for user interface and JDBC with a MySQL database for persistent storage. The project is developed in a NetBeans project and implements the DAO (Data Access Object) design pattern to separate data access logic from the presentation layer. The distribution includes the MySQL Connector/J 8.0.23 driver.

2. Background and Literature Review

2.1. Background Study

As lifestyle-related health problems grow, nutrition awareness is becoming increasingly important. Those who are aware of their calories and macro nutrient intake are more inclined to make good choices about what to eat. Desktop apps allow you to track your personal health in a safe offline environment without having to be online or pay for a service subscription.

The Nutrition Tracker is a simple database driven tool for recording food items, daily intake and allows users to evaluate their diet against self-defined goals for calories, protein, carbohydrates, fats, and fiber.

2.2. Literature Study

Food tracking systems typically include a food database, consumption logging, user authentication, and reporting. Relational database model allows to link food items, users, and log records with foreign key relationships. This allows for data consistency and cascade deletion. Health apps often use role-based access as a design pattern, letting users use the app for themselves or to manage the diet of a person under the supervision of a trainer or nutritionist. The DAO pattern is a common pattern in Java applications to abstract the persistence logic and make the application more testable and maintainable.

2.3. Existing System

Without a specialized tool, users generally track their nutrition using spreadsheets, paper logs, or generic mobile apps. These methods have limited validation, no integration between food definitions and daily logs and no administrative oversight. The Nutrition Tracker builds on this by storing all data in a MySQL relational database, enforcing referential integrity, separating user and admin roles, and providing a structured Swing GUI for all interactions.

3. Overall Description

Nutrition Tracker is a desktop application which is designed for two types of users, regular user, and administrator. The app then opens up to a login screen and then directs the authenticated user to either the User Dashboard or the Admin Dashboard depending on their role. All data is stored in a MySQL database named nutritiondb.

3.1. Product Perspective

The system is a desktop-based standalone application. It does not depend on any external web service. It uses JDBC to connect directly to a MySQL instance, either local or hosted on the network. The DatabaseConnection class is used to handle the singleton connection, and will auto-create the database, schema, and a default admin account on first run if it does not exist.

3.2. Product Features

User Module:

- Login to the system.
- Add new food items with nutritional values per 100g. Edit or delete existing entries.
- Select a food item and enter grams consumed for a selected date.
- View a summary of all nutrients consumed today compared to personal goals.
- Browse past food logs by date to review previous consumption.
- Read dietary notes sent by the admin/adviser.

Admin Module:

- Authenticate as an administrator using role-based session logic.
- Register account and set role.
- View all registered users and their account status.
- Inspect a selected user's food logs, nutrition history, and goals.
- Compose and send nutritional guidance or feedback notes to users.
- View and manage all notes sent across the system.
- Deactivate or delete user accounts as needed.

3.3. User Classes and Characteristics

Daily users are those who wish to record the amount of food they eat and how well they are eating based on their nutrition goals. To do this, they require a simple interface for entering food items, logging meals, and tracking their progress towards meeting nutrition goals through food management, logging, and summary display screens.

Administrators (including registered dietitians; trainers; support staff who manage customer accounts and provide nutritional advice) use the administration dashboard to review user profiles and send notes.

3.4. Operating Environment

The system has been developed in a desktop-based environment using Java and Java Swing for the user interface graphics and MySQL as the backend database system. It runs on Windows, macOS and Linux, if you have Java JDK 8 or newer installed. Database connectivity is done using MySQL Connector / J and project compilation and management is done using Apache Ant inside the development environment of NetBeans. The system is accessed through a graphical desktop interface on the local machine or network environment.

3.5. Design and Implementation Constraints

- The application depends on MySQL and a configured database connection.
- The DAO pattern is used throughout; all database operations go through interface-implementation pairs.
- The UI is built entirely with Java Swing (no JavaFX or third-party UI libraries).
- GroupLayout.jar library from NetBeans is required for certain form layouts.
- Passwords are currently stored as plain text; a hashing mechanism is planned for future versions.

3.6. Assumptions and Constraints

It is assumed that users have basic knowledge of using the application and that administrators understand task given to them. Additionally, it is assumed that database tables have been imported and the server environment is set up correctly before the application is utilized.

4. External Interface Requirements

4.1. User Interfaces

Application screens are implemented in Java Swing with the use of JFrame and JPanel components. These are Login Frame with stylized authentication fields, User and Admin Dashboards with side navigation, Add Food Frame to input nutritional information and Log Food Frame to track food consumption. Users can view daily and historical food logs in the Food Today and Food History Frames. Administrators have dedicated admin screens to managing users and reviewing individual logs and notes.

4.2. Hardware Interfaces

The application requires no special hardware-desktop or laptop computer with a keyboard and mouse is sufficient. Sufficient disk space is needed for Java runtime, the JAR file, and the MySQL database files. A network interface is required only if MySQL is hosted on a remote server.

4.3. Software Interfaces

The application interacts with several software components to support its functionality. It uses a MySQL database accessed through JDBC and uses PreparedStatement for all queries. MySQL Connector/J 8.0.23 and AbsoluteLayout are included in the project in dist/lib/ directory. It also uses the Java Standard Library, including javax.swing for the user interface, java.sql for database access and java.util for collections.

4.4. Interaction between Hardware and Software

Upon starting the application, the DatabaseConnection class opens a JDBC connection to MySQL and creates the nutritiondb database if it does not already exist, then runs DDL statements to create all the required tables and the default admin account. Subsequent user interactions, DAO method calls, perform parameterized SQL queries on the database. All UI events are processed on the Swing event dispatch thread; database calls are made from action listeners but are synchronous in the current implementation.

5. Other Requirements

5.1. Use Case Diagram for Nutrition Tracker System:

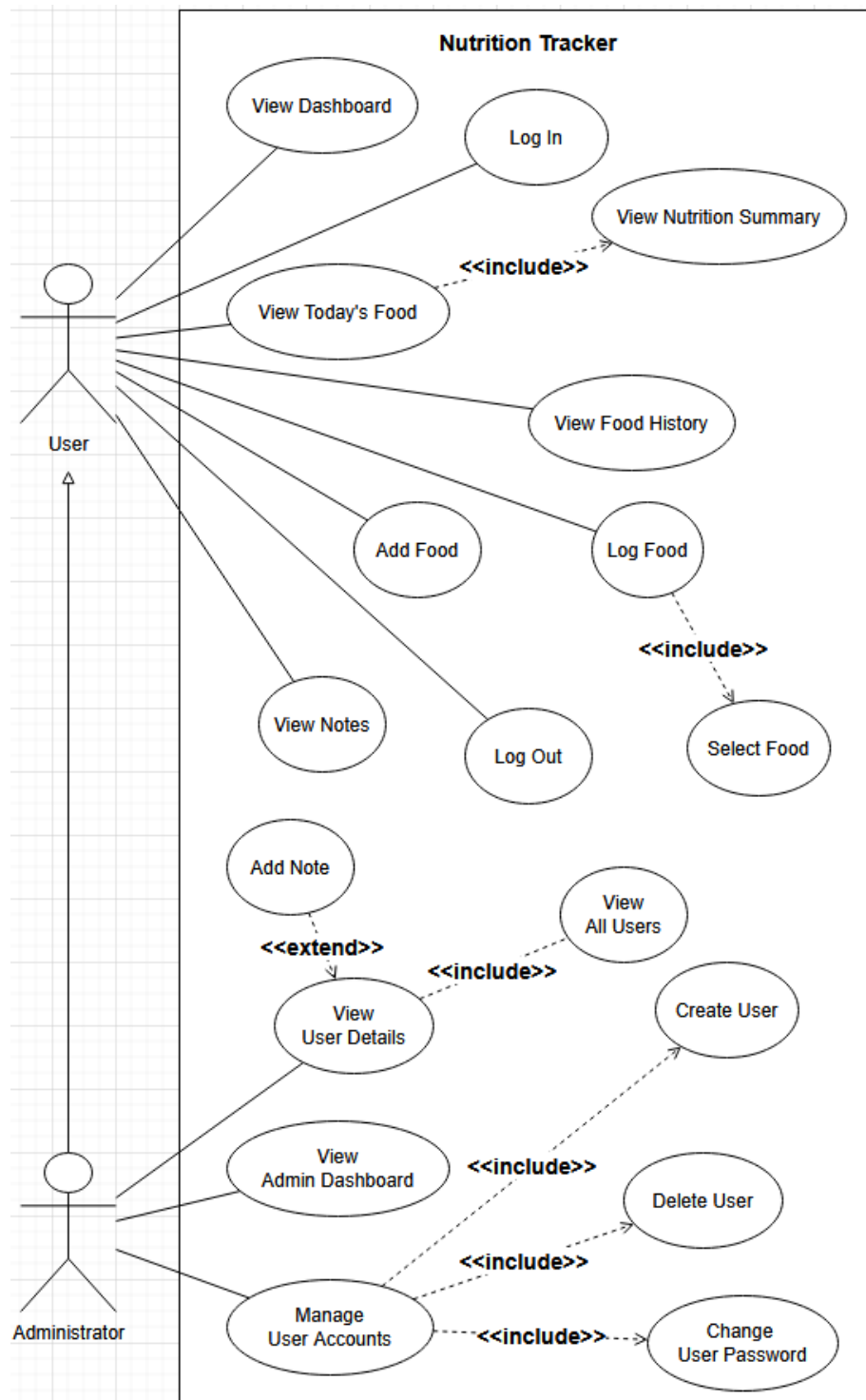


Figure 1: Use Case Diagram for Nutrition Tracker System

5.2. ER Diagram for Nutrition Tracker System:

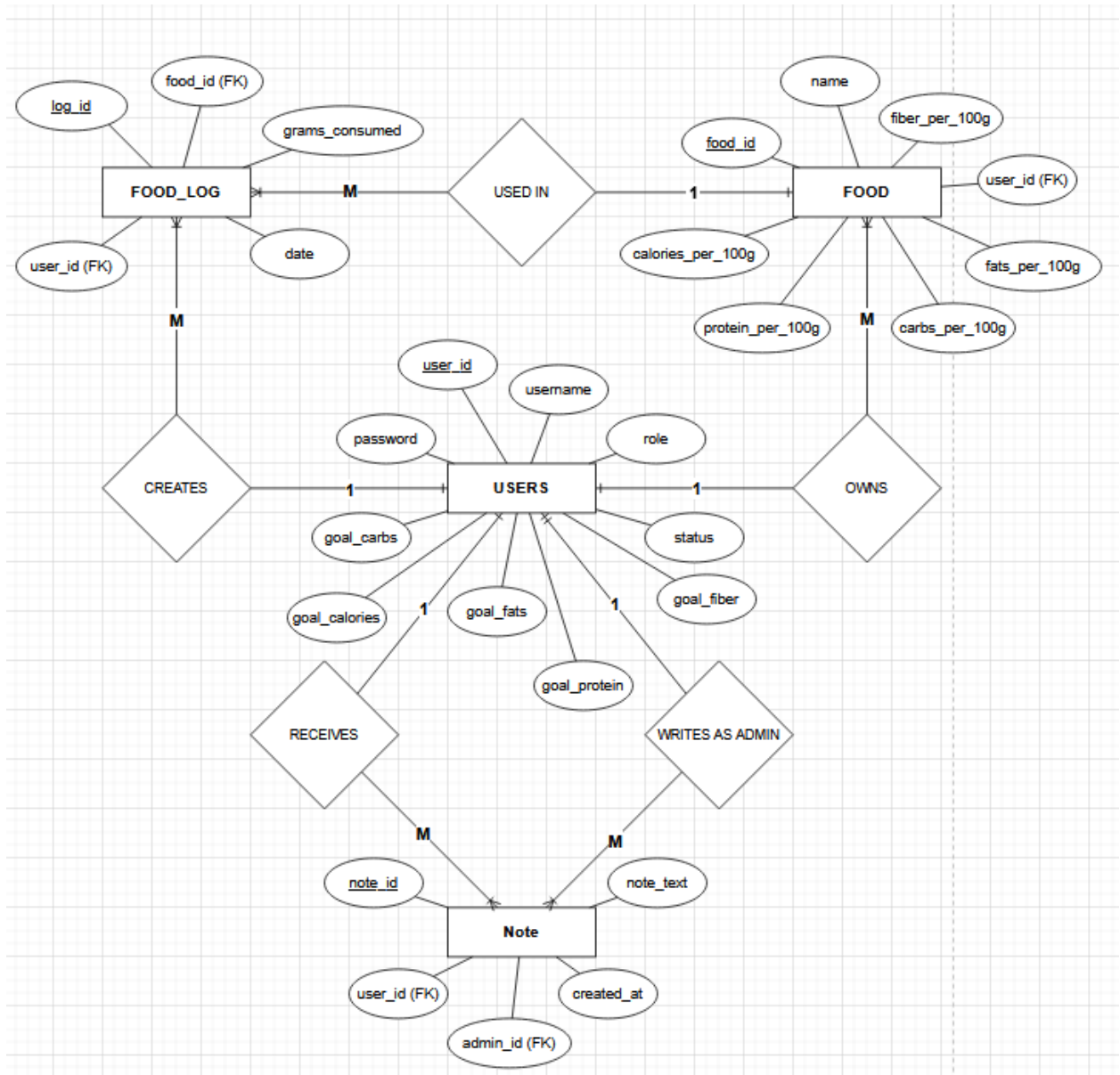


Figure 2: ER diagram for Nutrition Tracker System

5.3. Class Diagram for Nutrition Tracker System:

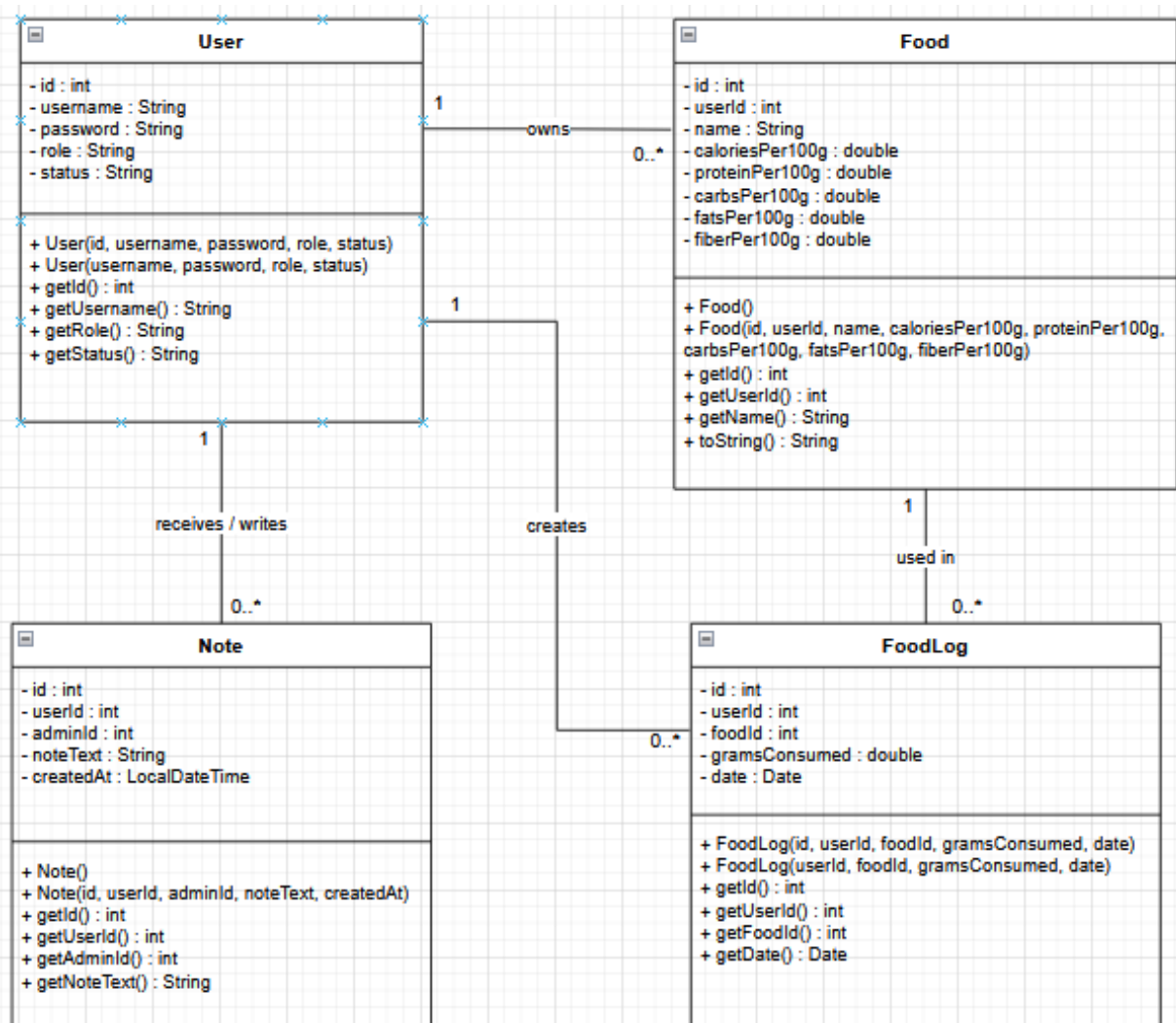


Figure 3: Class Diagram for Nutrition Tracker System

5.4. Data Flow Diagram for Nutrition Tracker System:

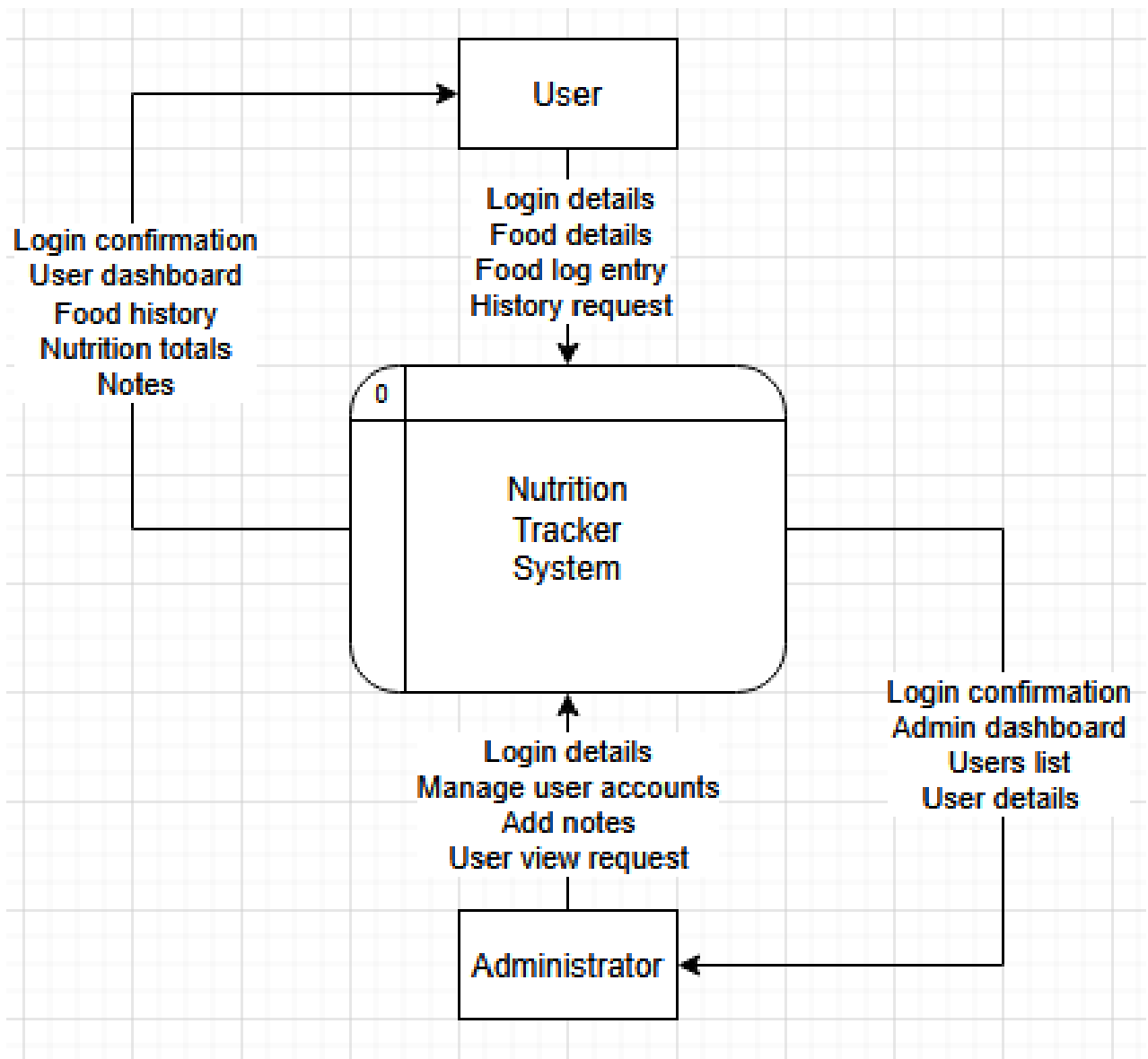


Figure 4: Level 0 DFD for Nutrition Tracker System

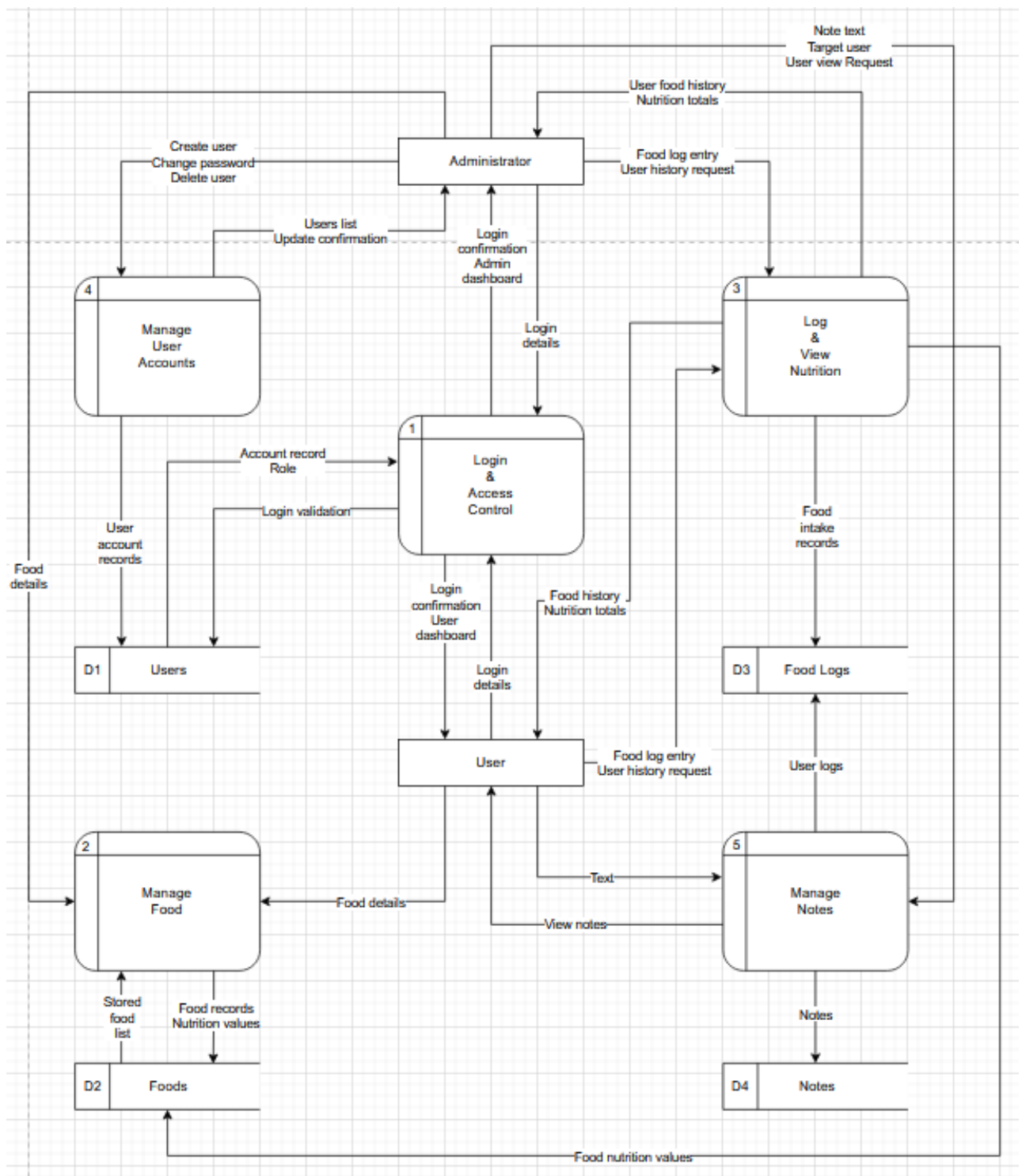


Figure 5: Level 1 DFD for Nutrition Tracker System

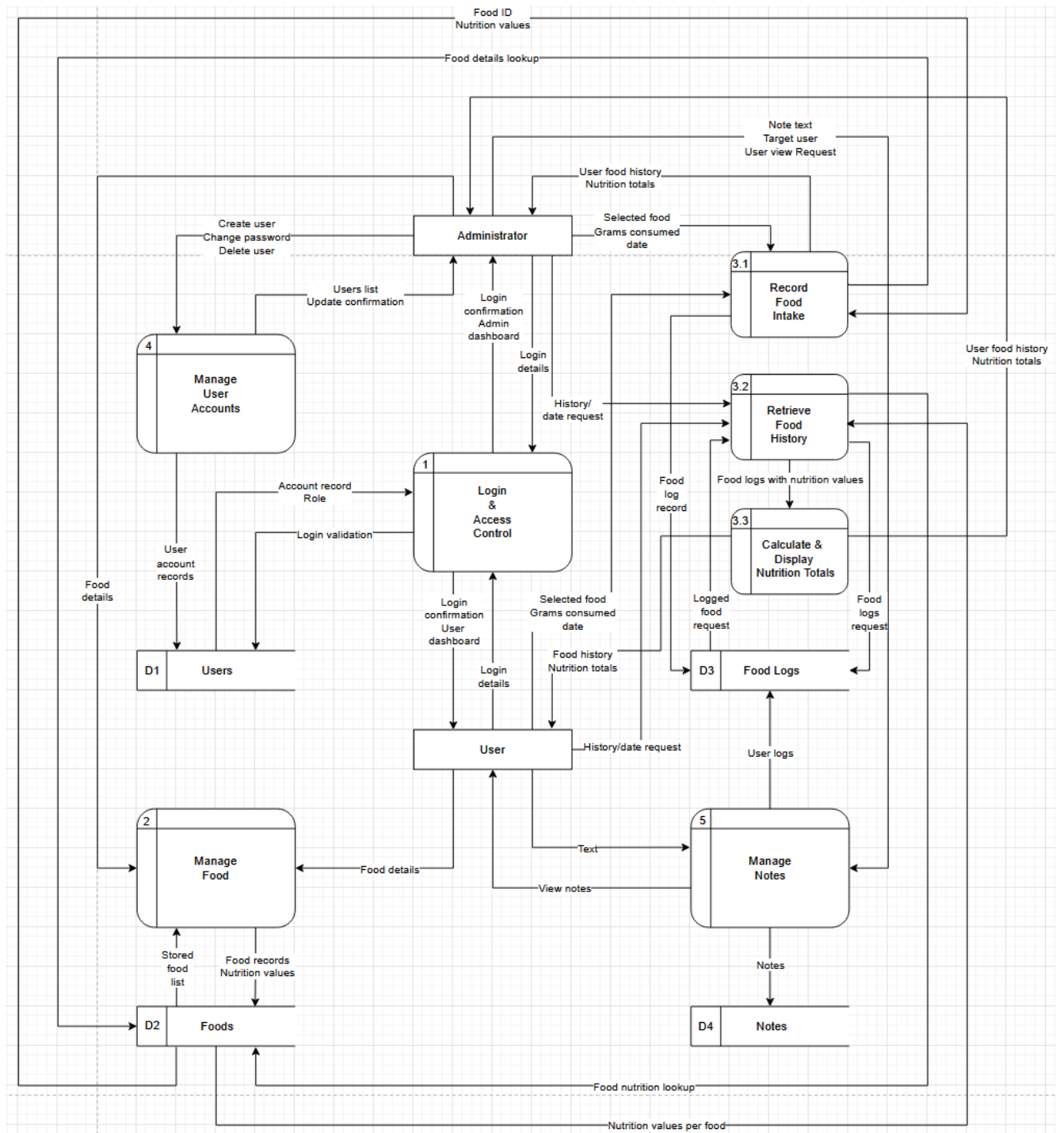


Figure 6: Level 2 DFD for Nutrition Tracker System

6. Non-Functional Requirements

6.1. Performance Requirements

- Login, dashboard launch, and food list load should complete within 2 seconds under normal database conditions.
- Nutrition summary calculation for a day's log should be computed in a single aggregated SQL query.
- The application should remain responsive during database operations; long-running queries should not freeze the UI.

6.2. Safety requirements

- The default admin account is created automatically on first run to ensure the system is always administrable.
- ON DELETE CASCADE constraints on food_log and notes prevent orphaned records when a user is deleted.
- A unique constraint on (user_id, name) in the foods table prevents duplicate food entries per user.

6.3. Security Requirements

- All SQL queries use JDBC PreparedStatements to prevent SQL injection.
- Role-based logic in the login flow ensures users cannot access the admin dashboard.
- Passwords should be stored using a secure hashing algorithm (e.g., BCrypt) in future versions; currently stored as plain text.
- Database credentials are configured in DatabaseConnection.java and should be protected through environment variables or a configuration file in production deployment.

6.4. Software Quality Attributes

- Usability: The interface should be easy for customers and administrators to understand.
- Maintainability: DAO interfaces decouple business logic from persistence, making it straightforward to switch databases or add new queries.
- Reliability: Foreign key constraints and UNIQUE indexes maintain referential integrity and prevent duplicate data.
- Scalability: The schema supports multiple users and large numbers of food and log entries without structural changes.
- Portability: Java and JDBC are cross-platform; the application runs on any OS with a JRE and MySQL access.

7. Gantt Chart

	1st week	2nd week	3rd week	4th week	5th week	6th week	7th week
Planning							
Analysis							
Design							
Coding							
Testing							
Implementation							
Documentation							

Figure 7: Gantt Chart for Nutrition Tracker System

8. Conclusion

The Nutrition Tracker is a functional and well-organized Java desktop application for tracking personal diet. It is designed to keep user and administrator functionality separate, is built on a normalized MySQL relational database for persistence and employs the DAO pattern for a clean separation of presentation from data access. Users can create and maintain their food catalog, log their daily food intake, and track their achievement of individualized nutrition goals. Administrators can monitor all users' usage and can provide diet advice to users using the notes feature.

The application illustrates many practical implementations of core Java fundamentals such as implementing a Swing GUI, using JDBC for database connectivity, following an object-oriented design, and utilizing interface-based abstraction. It provides a solid basis for subsequent expansion toward a health monitoring application that is ready for production.

9. Recommendations

- Implement password hashing using BCrypt or SHA-256 before any public or shared deployment.
- Move database credentials out of source code into an external configuration file or environment variables.
- Add input validation on all form fields with clear error messages to improve usability.
- Introduce asynchronous database calls (SwingWorker) to prevent UI freezing on slow queries.
- Add a search and filter capability to the food list and food history screens.
- Provide a visual nutrition chart (bar or pie) on the Today summary screen for easier interpretation.

10. Future Scope

Planned future improvements to the Nutrition Tracker include graphical reporting with daily, weekly, and monthly nutrition trend charts, and a barcode scanner interface that will allow for automatic population of food items through a public nutrition API. The system can also include a goal recommendation engine which recommends daily nutrition goals based on age, weight, height, and activity level. Other features include export of nutrition reports to PDF or CSV format for sharing with healthcare providers, a mobile companion application that synchronizes with desktop database, multi-language support for better accessibility, and two-factor authentication for admin accounts to improve security.